

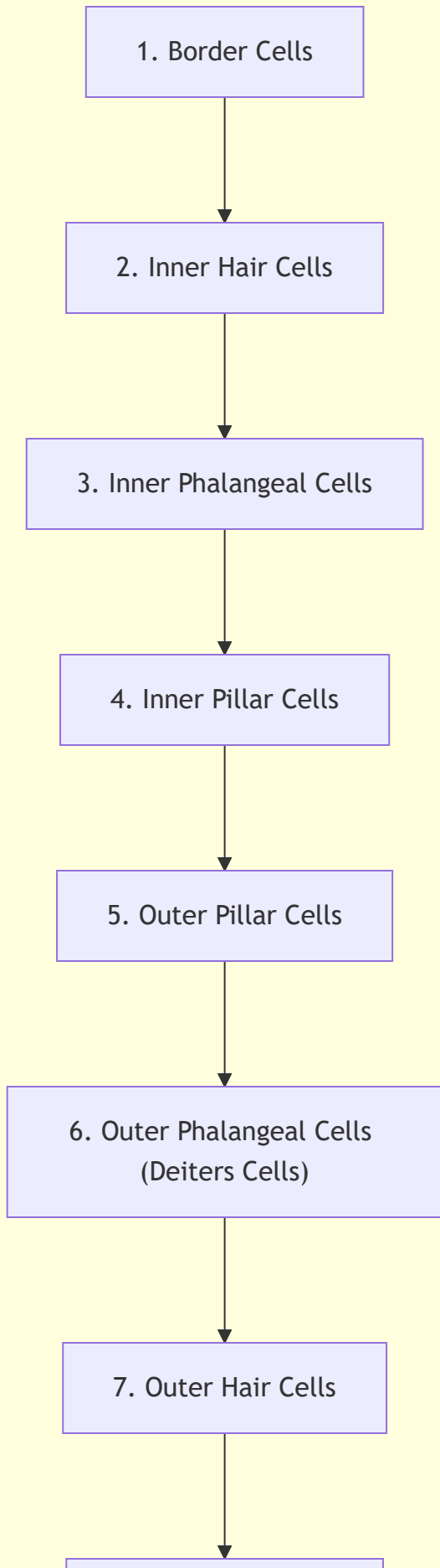
ORGAN OF CORTI

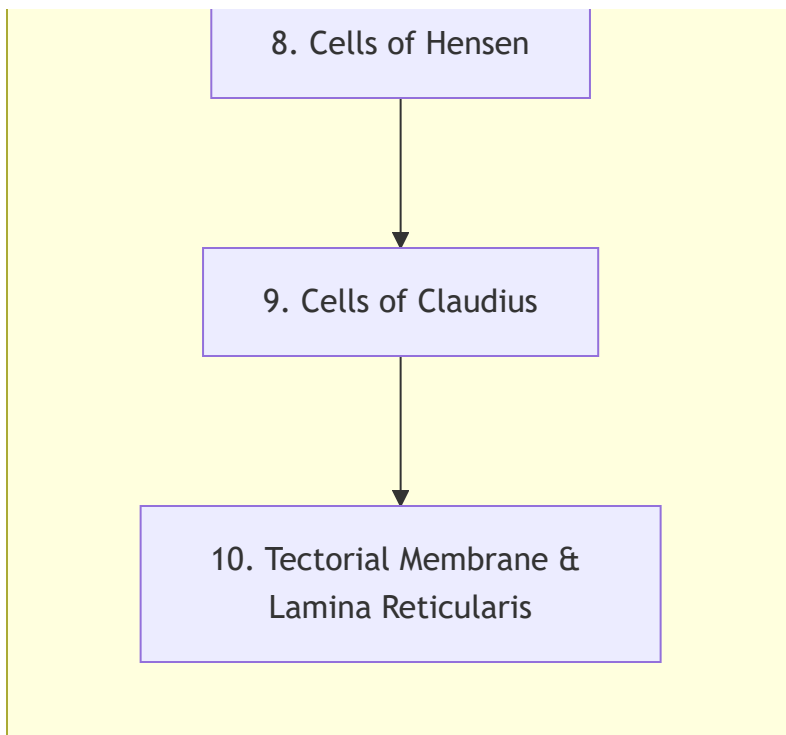
- **Definition :** The receptor organ for hearing; a specialized neuroepithelial structure situated in the cochlea.
- **Situation & Extent :**
 - Rests upon the lip of the **osseous spiral lamina** and the **basilar membrane**.
 - Extends throughout the length of the **cochlear duct**, except for a short distance at either end.
- **Roof :** Formed by the gelatinous **tectorial membrane**.

CELLULAR ARCHITECTURE

Made up of sensory elements called **hair cells** and various specialized **supporting cells**. The cellular elements are arranged in a specific medial-to-lateral order from the center (modiolus) toward the periphery:

Arrangement of Cells (From Center toward Periphery)





DETAILS OF CELLULAR COMPONENTS

1. Border Cells

- Slender columnar cells arranged in a single layer on the **tympanic lip** along the inner side of the inner hair cells.
- The apical surfaces of border cells possess a distinct **cuticle**.

2. Inner Hair Cells

- **Shape** : Flask-shaped cells, broader than outer hair cells.
- **Arrangement** : Arranged in a **single layer / row**, occupying only the upper portion of the epithelial layer.
- **Base** : The rounded base of each cell rests upon adjacent supporting cells termed **inner phalangeal cells**.
- **Apical Surface** : Bears a cuticular plate and specialized **stereocilia** (the largest stereocilium is termed the **kinocilium**).
- **Innervation** : Inner and outer hair cells collectively form the **receptor cells** of hearing; both receive afferent and efferent nerve fibers.

3. Inner Phalangeal Cells

- Act as primary supporting cells for the inner hair cells.
- Arranged in a single row along the inner surface of the inner pillar cells.
- **Base** : Rests directly on the **basilar membrane**.
- **Apex** : The cuticular plate of these cells resembles finger bones (**phalanges**).

4 & 5. Inner and Outer Pillar Cells (*Rods of Corti*)

- **Structure** : Each pillar cell consists of a broader base, an elongated body/pillar, and a head at the tip.
- **Attachments** :
- **Bases of inner pillar cells** : Lie close to the lip of the osseous spiral lamina (*tympanic lip*).
- **Bases of outer pillar cells** : Rest close to the **basilar membrane**.
- **Tunnel of Corti** : The inclined pillars meet at their heads to form a series of arches, enclosing a central triangular space termed the **Inner Tunnel / Tunnel of Corti**.

6. OUTER PHALANGEAL CELLS (*Cells of Deiters*)

- Supporting cells for the **outer hair cells**.
- Morphologically **tall columnar cells**.
- Send **phalangeal processes** upward between hair cells to form part of the **lamina reticularis**.
- **Space of Nuel** : Fluid-filled space located between the innermost outer phalangeal cells and outer pillar cells.

7. OUTER HAIR CELLS

- Columnar cells occupying the superficial portion of the epithelium of the organ of Corti.
- **Bases** are supported directly by the **outer phalangeal cells**.
- Internal structure is similar to that of **inner hair cells**.

8. CELLS OF HENSEN

- Tall columnar cells forming the **outer border cells** of the organ of Corti.
- Arranged in several rows on the **basilar membrane**, located laterally to the outer phalangeal cells.
- **Outer Tunnel** : The fluid space situated between the **outer phalangeal cells** and the **cells of Hensen**.

9. CELLS OF CLAUDIUS

- **Cuboidal** in nature; line the lower surface of the **external spiral sulcus**.
- **Boettcher Cells** : A group of specialized cells situated between the **cells of Claudius** and the **basilar membrane**.

10. TECTORIAL MEMBRANE & LAMINA RETICULARIS

- **Tectorial Membrane** :
 - Extends from the **vestibular lip** to the level of the **cells of Hensen** (forms the roof of the organ of Corti).
 - Movements and shear of the tectorial membrane by sound vibrations in the **endolymph** stimulate the stereocilia/processes of the hair cells.
- **Lamina Reticularis** :
 - Formed by the interlocking cuticular plates/processes of all supporting cells, providing a rigid apical framework.

STRUCTURAL SPACES & OUTER CELL SUMMARY

Component / Space	Morphology & Location	Key Functional Role / Relation
Outer Phalangeal Cells (Deiters)	Tall columnar supporting cells	Hold bases of outer hair cells; processes form lamina reticularis .

Component / Space	Morphology & Location	Key Functional Role / Relation
Space of Nuel	Fluid space	Located between innermost outer phalangeal cells and outer pillar cells.
Outer Hair Cells	Columnar sensory cells in 3–4 rows	Auditory electromotility / cochlear amplification.
Cells of Hensen	Tall columnar cells in several rows	Form lateral outer border of organ of Corti.
Outer Tunnel	Fluid-filled space	Located between outer phalangeal cells and cells of Hensen.
Cells of Claudius & Boettcher	Cuboidal cells over spiral sulcus	Provide peripheral epithelial support on the basilar membrane.
Tectorial Membrane	Acellular gelatinous membrane	Overlies hair cells; shears stereocilia during endolymph movement.